



COUNTING FROM THE FIRST TICK

Life, the Universe, and God

A Software Engineer's Instrument for the Immeasurable

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2026 · built against the ucal source tree

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Before we begin

Three things are true about this project, and the third only follows if you accept the first two. So they go here, on the first page, before anything has a chance to soften them.

The artifact is real. `ucal` is working Rust — six crates, a library and a command line, published and installable. Time is an unsigned integer count of Planck-time units since a stipulated datum. There is no floating-point value anywhere in the workspace, not in a signature, a field, an intermediate, or the rendering path. Every constant in it is reproducible by two independent derivations that agree bit for bit. At the commit this book is pinned to, 381 tests pass on both integer backends. You can check every sentence in that paragraph; the last page of this book tells you how.

Its practical utility is questionable. No task you have today needs a Planck-tick count since a stipulated datum. It does not compete with `chrono`, `time`, or `hifitime`, and it does not try to. Nothing on its ladder of units is near a second or an hour — a second is 21.4 beats, an hour is 24.6 arcs — so it is not merely unnecessary for ordinary work but actively awkward for it. There is exactly one qualification to this, it concerns timekeeping across more than one planet, it takes up a single page in Part V, and it is not a rescue.

Therefore it is research of another kind. Art, philosophy, theology — conducted in the medium of a working program. That is not a consolation prize awarded after usefulness failed. It is what the thing is.

The rigor is the medium

Here is the part that matters, and it is easy to misread as modesty.

That the code compiles, that the constants reproduce, that the tests pass — these are not preliminaries to the argument. They **are** the argument, in the same way that the metre of a poem is not preliminary to the poem.

An essay can assert that a distinction ought to be respected. It can argue at length, and persuade you, and then it is over, and the discipline it recommended

survives exactly as long as the next author's attention does. A type system can do something an essay cannot: it can make violating the distinction **fail to build**. Not discouraged. Not deprecated with a warning. Refused, by a machine, to a person who disagrees.

That difference is the thesis of this book, and the reason it is a book about a program rather than a program with a book attached.

What this is not

Some of these will be obvious. They are listed because a book that surveys nine philosophical and religious traditions alongside a Rust crate will be misread in predictable directions, and it is cheaper to close those doors now than to keep apologising later.

Not an apologetic No tradition is argued true here. The code is evidence for none of them. Where a tradition and the instrument agree, that is reported as a convergence and never as a vindication of either.

Not a proof of usefulness The second fact above stands unretracted.

Not a claim of discovery When a tenth-century argument turns out to match a twenty-first-century type signature, the older text is not thereby shown to have anticipated anything. Convergence is not anticipation.

Not numerology No constant, base, or magnitude in this system acquires meaning by resembling a number in a tradition. Base 5 was chosen because $5^5 = 3125$ gives a tier of exactly five digits. That is the whole reason. The book says so plainly and then reports, as a curiosity and not as evidence, that the first place the research went looking for a five, it found one waiting.

Not a tutorial If you want to use the crate, its documentation is better at that than this book will be.

Not a defence of the datum Tick zero is stipulated. That it is stipulated is the point, and Part IV is four chapters about why.

Two paths through the book

The book serves two readers who are rarely the same person, and it does not ask either of them to pretend to be the other.

The **engineer's path** is Parts I, II, III, and V. It is a complete technical article about an unusual piece of software: what it computes, what its type system refuses, where the design lost arguments with the compiler, and how far the approach generalises beyond Earth. You can stop at the end of Part V and have received the whole of that.

The **reader's path** is Parts I, IV, VI, VII, and VIII. It is a book about what it means to build a measuring instrument that points at something it declares itself unable to measure, and about nine traditions that have thought carefully about origins, number, and time — read here as readers of the artifact rather than as authorities over it.

Part I belongs to both. Nothing later in the book may redefine what Part I establishes.

A NOTE ON MARKING

This book marks its own claims. Where a passage is **interpretation** — the author reading a meaning into the artifact — it sits in a ruled block that says so. Where a coincidence is genuinely striking but proves nothing, it is labelled **resonance**, and it may never appear as a premise for anything.

That apparatus is not decoration. It is the book submitting to the discipline it documents: the software marks where it stops computing, and the book marks where it stops asserting fact. If you delete every marked block, every technical claim in Parts I through III and V must still stand. Making sure that is true is a scheduled step in producing this book, not a hope about it.

A word about being wrong

The most valuable document in this project's history is a correction.

Two revisions of the specification asserted that the continued-fraction machinery reproduces the Julian and Gregorian calendar rules as convergents. The Julian rule does appear — it is the first convergent. The Gregorian rule does not appear at any depth, and a rule twelve times simpler is more accurate than it. The machinery contradicted its author, in writing, on a claim he had published twice.

That correction is in this book, dated, including the revisions that were wrong. It is in here because a system built to make its author's conclusions provable would

never have produced it, and because the charge that such systems are **always** built that way is nine hundred years old and deserves an answer made of evidence rather than assurance.

If you find yourself reading a passage that seems to be presenting that correction as foresight rather than as an error published — a demonstration of rigour rather than an argument the author lost — then the book has failed at the thing it cares most about, and you should say so.

PART I

Foundations

*Definitional. Nothing later in the book may redefine these,
and a reader who stops here must not leave with a false
impression.*

CHAPTER 1

1

Intent

This began as an irritation, and it is worth being honest that it began as an engineer's irritation rather than a philosopher's question.

Read almost any account of the early universe and you will find a sentence like this one: **recombination occurred about 380,000 years after the Big Bang**. It is a good sentence. It is doing real work. And if you stop and ask what its units are, something uncomfortable happens.

A year is the time Earth takes to go around the Sun. Not approximately — definitionally. The Julian year used in astronomy is exactly 365.25 days of exactly 86,400 seconds, and those numbers are what they are because of how fast one particular

rock spins and how long it takes to circle one particular star. So **380,000 years after the Big Bang** means $380,000 \times 365.25 \times 86,400$ seconds, which means it is a quantity expressed in units defined by the rotation and orbit of a planet that would not exist for another nine billion years.

The number is correct. It is not wrong in any way that matters to a cosmologist. But it carries a passenger, and the passenger is Earth.

The irritation, stated precisely

The problem is not that the units are arbitrary. All units are arbitrary. The metre is a bar in a vault, then a wavelength, then a fraction of a light-second; none of that makes it a bad metre.

The problem is **provenance leaking into arithmetic**. When you compute with Earth years, Earth's orbital period is inside every intermediate value. Usually that costs nothing. Occasionally it costs you a rounding you did not intend, because 365.25 days is not a whole number of anything and the conversions do not close. And structurally — which is the part that would not let go — it means the description of an event 13.8 billion years old is phrased in terms of a body that had no bearing on it.

WHAT THIS IS NOT AN ARGUMENT FOR

None of this is a criticism of how cosmology is done. Astronomers use Julian years because they are stable, agreed, and unambiguous, which is exactly what a unit should be. The complaint here is narrow and it is aesthetic before it is technical: a quantity ought to be expressible in units that do not smuggle in a planet.

What would have to be true instead

Suppose you wanted a time system with no Earth content in its arithmetic. What would it need?

It would need a **unit** that is not defined by any body's motion — something built out of constants of nature rather than out of a rotation.

It would need an **origin** that is not a calendar event, since every calendar epoch is someone's civil history.

It would need a **notation** in which writing fewer digits means saying less precisely, rather than meaning zero — because at these magnitudes you will constantly be quoting numbers you do not know to full precision, and a system that silently pads them with zeros is lying on your behalf.

And it would need to be honest about the one thing it cannot do: it cannot tell you what time it is on Earth without, at some point, being told about Earth. That contact has to happen somewhere. The question is whether it happens **in the arithmetic** or at a declared boundary you can point to.

That last question turned out to be the whole design.

What was built

ucal is the answer to those four requirements, and Part II is the technical account of it. In one paragraph: absolute time is an unsigned integer count of Planck-time units since a stipulated datum; those ticks are grouped in a ladder of powers of five, so a timestamp is the tick count written in base 5 and truncating it **is** rounding it; Earth enters through exactly one declared constant and leaves at the formatting boundary; and no floating-point value appears anywhere in the workspace.

It works. It is also, as the preface said and will keep saying, of questionable use.

Where the trouble started

The requirement that has no clean answer is the second one: the origin.

If time is a count since some zero, the zero has to be somewhere. And the obvious place to put it — the beginning — is not available, for reasons that took four chapters of this book to state properly and that Part IV is entirely about.

The short version, so you can carry it through Parts II and III: **the age of the universe is a measurement, and measurements have error bars, and an error bar cannot be the origin of an exact integer count.** The published figure is 13.787 billion years give or take 0.020 billion. That uncertainty, converted to ticks, is a number with fifty-eight digits. If the datum inherited it, every timestamp in the

system would inherit it too, and the exactness the whole design exists to protect would be theatre.

So the datum is **stipulated**. Declared, not discovered. Tick zero is a reference point that the specification says, in as many words, is not a measurement and not an observed event.

And then the interesting problem: how do you say that in a way that a future maintainer — or the author, three years later, in a hurry — cannot quietly forget?

INTERPRETATION

The answer this project arrived at is the reason it turned into a book. The physical claim about the origin is declared, cited, and given an exact magnitude. And then it is made **impossible to compute with**: a type with no arithmetic operations at all, guarded by a test that fails to compile if the type ever reaches an operand position.

You cannot forget a discipline that the compiler enforces. That is a small technical fact with a large philosophical shadow, and the shadow is what Parts IV, VII, and VIII are about.

What this book will and will not claim

Before Part II starts and the prose becomes technical for a hundred pages, here is the calibration.

It **will** claim that a distinction enforced in a type system is a philosophical contribution and not merely an implementation detail. It will claim that the approach generalises to any celestial body, and it will spend a chapter on exactly where it fails. It will claim that the instrument does philosophical work no prose argument can do, and it will demonstrate that rather than assert it.

It will **not** claim that time began, that any tradition is correct, that the base is meaningful, or that anyone should adopt this calendar.

WHAT THIS CHAPTER ESTABLISHED

- Cosmic durations are conventionally expressed in units defined by one planet's motion — correct, useful, and carrying a passenger.
- A time system with no Earth content in its arithmetic needs a physical unit, a non-calendrical origin, a notation where truncation means imprecision, and one declared point of contact with Earth.
- The origin cannot be measured, because a measurement has error bars and an exact integer count cannot inherit one.
- So it is stipulated — and the project's contribution is making that stipulation mechanically impossible to forget.

CHAPTER 2

2

The tick

Everything this system computes is an unsigned integer count of ticks. Nothing else is primitive — not the second, not the beat, not the day. If you understand why the tick was chosen and, more importantly, what choosing it does **not** claim, you have the foundation of the whole instrument.

TERM Tick

The Planck time, $t_P = \sqrt{\hbar G/c^5} \approx 5.391 \times 10^{-44}$ seconds. In this system it is the atomic unit of duration: the smallest quantity representable, and the unit in which every other quantity is counted.

Why this unit

The Planck time is composed from three constants: the gravitational constant G , the reduced Planck constant $\hbar$, and the speed of light c . Those three bound gravitation, quantum action, and the propagation of causality respectively — and there is exactly one combination of them with the dimension of time.

That is the appeal, and it is worth stating carefully because it is easy to overstate. The tick is not defined by any body's motion. It does not know about Earth, or about the Solar System, or about matter being organised into planets at all. It is the interval you get when you ask the three limiting constants of physics what a duration would be, and they answer.

For a system whose entire purpose is to have no Earth content in its arithmetic, that is the right floor.

What the tick is not

Here is where a great deal of writing about the Planck time goes wrong, and where this project declines to follow.

The tick is not a quantum of time. It is the resolution floor of an instrument, not a structure discovered in the world. Nothing in this project asserts that time comes in discrete lumps, that there is a smallest possible interval, or that asking about a shorter duration is physically meaningless. The system cannot represent a shorter duration; that is a fact about the system.

The distinction matters more than it looks, because asserting otherwise would silently take a side in an argument that has been running for twenty-four centuries.

Aristotle holds in **Physics** VI that time is a continuum: divisible without limit, and the instant is a **limit** of an interval rather than a part of it, in the way that a

point is a limit of a line and not a very short line. On this view a smallest duration is not merely unobserved but incoherent.

Against that, temporal atomism has a long and respectable history. Epicurus held that there are **minima** of time as there are minima of magnitude. The Kalām tradition developed a thoroughgoing atomism in which time is composed of indivisible moments and accidents do not endure from one to the next.

The instrument is discrete. If discreteness were asserted as a claim about the world, this project would have joined Epicurus and the Kalām against Aristotle — and it would have done so not on the strength of an argument but as a side effect of choosing an integer type.

That is not a respectable way to take a metaphysical position. So the position is declined, explicitly, and the declining is written down: the tick is where the instrument's resolution stops, and the instrument makes no claim about where the world's does.

A RULE YOU WILL SEE AGAIN

This pattern — **the implementation forces a choice; name the choice; refuse the claim that would come free with it** — recurs throughout the design. It is the same move that Part IV makes about the datum, at much greater length and with much higher stakes.

The concession: the tick's length

There is one place where the tick is not free of Earth, and it should be admitted here rather than discovered later.

The Planck time's **numerical value** is known only as well as G is measured, and G is the worst-measured of the fundamental constants by a wide margin. Worse, expressing it at all requires a unit of time to express it **in** — and that unit is the SI second, which is defined by a caesium transition counted by instruments on Earth.

So the tick's length is fixed by convention against the second. The system declares a value, records where the value came from, and uses that declared value everywhere. It does not re-derive it, and it does not pretend the value is more certain than the measurement behind it.

INTERPRETATION

What this concedes is **metrology** and nothing else. The arithmetic contains no Earth content: ticks are counted, not converted. What is Earth-flavoured is the statement of how long a tick is in seconds — which is a sentence about translating between two systems, not a fact used inside either.

Whether that is a satisfying answer is a fair question, and Part III returns to it. It is at least an honest one, and its honesty is structural: the conversion lives at a declared boundary you can point to rather than dissolved into the operations.

Counting, not measuring

One more consequence, which will matter in Part IV.

Because time here is a **count**, the system's exactness is the exactness of integer arithmetic. Tick 8,070,204,002,895,596,515,944,343,085,635,637,180,530,466,139,316,558,837,890,625 is a specific tick, and adding one to it gives the next one, and that is all perfectly exact — in the same way that the number of steps you have walked is exact whether or not you know it.

What is **not** exact, and cannot be made exact by any amount of careful arithmetic, is the correspondence between tick zero and any physical event. That is a separate question, it has a separate answer, and keeping those two things apart is the discipline the rest of this book is about.

WHAT THIS CHAPTER ESTABLISHED

- The tick is the Planck time, built from G , $\hbar$, and c — the one combination of the three limiting constants with the dimension of time.
- It is the resolution floor of an instrument, **not** a quantum of time. Asserting otherwise would take a side in a live argument as a side effect of choosing an integer type.
- Its length in seconds is fixed by convention against SI. This concedes metrology and nothing else, at a boundary you can point to.
- Time is counted, not measured. The count is exact; what tick zero corresponds to is a different question entirely.

CHAPTER 3

3

The zero of time

A count needs somewhere to start counting from. This chapter says where, and says the one thing about it you need in order to read Parts II and III without being misled. The full argument is Part IV, four chapters of it, and the deferral is deliberate — but a reader who stops at the end of Part I must not leave with a false impression, so the essential point is made here and made plainly.

TERM Datum

Tick zero. A **stipulated** reference point, conventionally identified with the Friedmann–Lemaître–Robertson–Walker $t \rightarrow 0$ limit. It is not a measurement and not an observed event.

The domain is unsigned

Absolute time in this system is an unsigned integer. There is no tick -1 .

This is not a storage optimisation. It is a statement about what the system is willing to represent: nothing precedes the datum, because the datum is where the count begins, and a count does not go backwards past its own start.

Ask for an instant before tick zero and you do not get a negative number. You get an error — **UCAL-E0020** — and the operation fails.

INTERPRETATION

There is a difference between answering a question with a negative number and refusing the question, and the system takes the second option deliberately.

A negative tick count would be an **answer**: it would say that the time before the datum exists, is measurable, and happens to lie on the other side of a chosen origin — the way 1 BC lies on the other side of the Gregorian epoch. The refusal says something else: that the question, as posed to this instrument, is malformed.

Augustine gives the classic form of the move in **Confessions** XI, answering what God was doing before creating heaven and earth. His answer is not a description of that interval but a rejection of the question’s shape, because time is among the things created and there is no “before” of the sort the question wants. Whether or not you find the theology congenial, the logical structure is exactly **UCAL-E0020**: an error, not a negative number.

Why it is stipulated and not measured

Here is the one-line version. Part IV gives it four chapters and three further reasons; this is the one you can carry.

Exactness cannot come from measurement.

The age of the universe is a measured quantity. The current best figure is 13.787 billion years, plus or minus 0.020 billion. That is an excellent measurement. It is also, like every measurement, an interval rather than a point.

Convert that uncertainty into ticks and it becomes:

- ● ● *ucal datum* — the claim's magnitude

[illegible]

Fifty-eight digits. About 0.145% of the whole span from the datum to now.

Now suppose the datum were defined as the measured age. Every timestamp in the system would inherit that uncertainty, because every timestamp is an offset from the datum. The exact integer arithmetic — the thing this entire design exists to protect — would be computing precise-looking answers on top of a foundation that wobbles by a number with fifty-eight digits in it.

You cannot get an exact origin from an inexact measurement. So the origin is not taken from the measurement. It is **declared**, and the measurement is recorded separately.

What is actually claimed

Precision about this matters, so here is the claim in full, with nothing left implicit.

The system asserts that tick zero is tick zero. That is a definition, and definitions are not the sort of thing that can be wrong.

It asserts, **separately and as metadata**, that tick zero is conventionally identified with the FLRW $t \rightarrow 0$ limit, and that the published uncertainty in that identification is ± 0.020 Gyr, citing Planck 2018. This is a claim about the world and it may be revised.

And it asserts nothing whatever about whether time began, whether there was a first moment, or whether the FLRW limit corresponds to a physical event at all.

DEFERRAL IS NOT EVASION

You have now been told that the datum is a decision rather than a discovery, and why the short reason forces it. What Part IV adds is three further reasons the specification gives, one more that it does not, and — the part that turned this project into a book — what was **done** about it: how a claim can be declared, cited, given an exact magnitude, and simultaneously made impossible to compute with.

If you are on the engineer’s path and skipping Part IV, the sentence to carry forward is this one: the uncertainty in the age of the universe does not make the timestamps uncertain, because it is not an operand.

The datum is in ordinary company

One last thing, because “stipulated origin” can sound exotic and is not.

TAI’s zero is 1 January 1958, chosen by agreement. The Julian Day epoch is 1 January 4713 BC, chosen because it made a convenient common multiple of three cycles. The Unix epoch is 1 January 1970, chosen because it was recent and round. None of these is a discovery about the universe; all of them are decisions that turned out to be useful.

INTERPRETATION

What is unusual here is not that the origin is stipulated. It is that the stipulation is **declared as such in the artifact**, with the physical claim held separately and marked non-computable — rather than being a fact about the system’s history that you would have to go and read about.

Most epochs are stipulated and silent about it. This one is stipulated and says so, in a place the compiler can reach.

WHAT THIS CHAPTER ESTABLISHED

- Tick zero is the datum: stipulated, conventionally identified with the FLRW $t \rightarrow 0$ limit, and explicitly not a measurement.
- The domain is unsigned. A request for a time before the datum is an error, not a negative number — a refusal of the question rather than an answer to it.
- Exactness cannot come from measurement: the ± 0.020 Gyr uncertainty is a 58-digit number of ticks, and an exact count cannot inherit it.
- The physical claim is recorded separately as metadata, cited, and — as Part IV shows — made impossible to use in arithmetic.

CHAPTER 4

4

The universe second

A tick is far too small to think in. It is 5.4×10^{-44} seconds; the age of the universe is about 8×10^{60} of them. Nobody reads a 61-digit number.

So there is a ladder of larger units above the tick, and it has an unusual property: every rung is a whole power of five of the one below, and the whole ladder is a whole power of five of the tick. That property is what makes the notation work, and this chapter is about what it buys and what it costs.

TERM Beat

5^{60} ticks, about 46.762 milliseconds. The reference rung of the ladder, and what the specification calls the **universe second** — a unit of human-noticeable size with no Earth content whatever.

Why base five

Because $5^5 = 3125$, and 3125 is a number of exactly five base-5 digits.

That is the entire reason. It is worth being blunt about it, because a book that spends its second half among Pythagoreans and Neoplatonists has an obligation to say clearly where its numbers came from, and this one came from a digit-packing convenience.

RULE N

No constant, base, or magnitude in this system acquires significance by resembling a number in a tradition. Not the five, not the sixty in 5^{60} , not the 3125.

The research did, as it happens, go looking. The first place it looked — Losev on the Neoplatonic tetrad — had a five waiting, and the Pythagorean tetraktys sums to ten and builds from four. Those are interesting facts about those texts. They are not evidence about this software, they were not inputs to it, and the book reports them once, in Part VI, labelled as what they are.

The ladder

Each rung is $5^5 = 3125$ of the rung below, which is to say each rung is exactly five base-5 digits wide. Rung k is 5^{60+5k} ticks, indexed from the beat at $k = 0$.

coarser			
	T5	<i>deep</i>	441.6 Myr
	T4	<i>drift</i>	141.3 kyr
	T3	<i>span</i>	45.2 yr
	T2	<i>sweep</i>	5.285 d
	T1	<i>arc</i>	146.1 s
	T0	<i>beat</i>	46.762 ms
	T-1	<i>flicker</i>	14.96 μ s
	T-2	<i>glint</i>	4.79 ns
finer			

Figure 2 — The named rungs of the tier ladder. Each is 3125 of the one below. The ladder continues in both directions; only the named tiers are shown.

What the ladder buys

Three things, and they are all consequences of the same fact — that a timestamp is the tick count written in base 5 and grouped in fives.

Truncation is rounding. Write fewer groups and you have said the same thing less precisely. There is no separate rounding step, no scaling, no loss of a different kind: the digits you dropped are exactly the precision you gave up. This is the property that makes the uncertainty discipline of Part II possible at all.

Prefix comparison is chronological comparison. Two timestamps compare in the order their digits compare, because base-5 positional notation is monotone. Sorting the text sorts the times.

Writing all the digits pinpoints one tick. There is no accumulated conversion error between the coarse view and the fine one, because they are the same integer read at different widths.

Here is one real instant, at full precision and then at three coarser tiers:

• • • *ucal explain — one instant, four precisions*

```

ticks
8070205189128471254993117657693008777530466139316558837890625

human      UC1 0031·0687·2481·3000·1638·3018:0779·2671·2006·1837·...
tiers:
  T5 deep   31
  T4 drift  687
  T3 span   2481
  T2 sweep  3000
  T1 arc    1638
  T0 beat   3018

```

The **0031** at the front is deeps since the datum. The **3018** before the colon is beats within the current arc. Every group is a base-5 number written in decimal, which is why none of them exceeds 3124.

INTERPRETATION

There is something worth noticing about that colon. It sits at the beat — at 5^{60} — and it is the only place in the notation where a human convenience has been allowed in. It is there because the beat is roughly the scale at which a person can notice a duration, so it is the natural place to put the decimal point of a time system built for beings who perceive at that scale.

That is a concession to the reader, not to Earth. It changes no arithmetic.

What the ladder costs

Nothing on this ladder is near anything you know.

A second is 21.385 beats — not a whole number, and not close to one. An hour is 24.6 arcs. A day is 0.189 sweeps. A year is 0.699 spans.

THE TWO SECONDS DO NOT DIVIDE

This is the sharpest form of the cost, and the system says so on its own ladder output:

The two seconds are incommensurable above T-6. One bridge second is 21.385061835 beats, not a whole number, because **BEAT** carries 5^{60} while **SECOND** carries only 5^{30} . They share a common measure only at the tick.

So the beat is not a second in disguise, and no amount of rescaling would make it one. They agree at the tick and nowhere above it — which is exactly why the tick is primitive rather than either of them.

That is the honest consequence of leaving the Earth paradigm. If you want units with no planetary content, you do not get to keep the hour. The hour **is** planetary content.

The mark

The project's own mark is a diagram of this chapter, and it is worth reading as one.

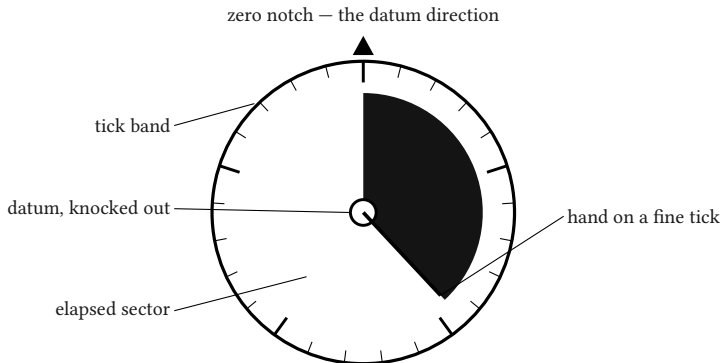


Figure 4 — *The mark*, read as a diagram. Five heavy divisions on the tick band, not twelve — the base is visible at a glance. The centre is negative space.

The centre dot has its core knocked out. That is deliberate: the datum is stipulated rather than observed, so the mark declines to draw anything there. The hand lands on a fine tick rather than a coarse division, because the claim the instrument makes is that it can pinpoint an arbitrary exact instant — not that it can round to a tidy one.

INTERPRETATION

A mark that showed a filled centre would be making a claim the specification refuses. The identity was drawn after the rule existed, and it obeys it. That is a small thing, and it is the sort of small thing this project is made of.

WHAT THIS CHAPTER ESTABLISHED

- The beat is 5^{60} ticks ≈ 46.762 ms — the **universe second**, a human-scale unit with no Earth content.
- Base five because $5^5 = 3125$ is five base-5 digits wide. Rule N: no further meaning, and the book says so before Part VI can be misread.
- The uniform ladder makes truncation **be** rounding, prefix comparison **be** chronological comparison, and full precision pinpoint one tick.
- The cost is that nothing on the ladder is near a second or an hour — one second is 21.385 beats, and the two share a measure only at the tick.

PART II

What was built

*An engineer may stop at the end of this part and have
received a complete technical article.*

CHAPTER 5

5

The domain

Part I established what is counted. This part is about how the counting is implemented, and it starts with the number that holds the count.

Three properties, and the third is the one that turns out to have consequences all the way into the cosmology: the domain is **unsigned**, it is **closed**, and every operation on it is **checked**.

Unsigned, and why that is not a saving

There is no tick -1 . Chapter 3 said why in one line — nothing precedes the datum, because the datum is where the count begins — and the type system is where that line is enforced.

`Instant` has no `Sub` implementation returning a signed value. Subtracting a later instant from an earlier one does not produce a negative `Delta`; it produces `UCAL-E0020` and the operation fails. To get the distance between two instants regardless of order you must ask for it by name, and the name says what you are doing.

THE SAME DISCIPLINE, APPLIED TO RATIONALS

`Ratio` — the exact rational type used throughout the derivations — carries the identical rule. `sub` refuses a negative result and directs the caller to `abs_diff`. That is Rule Z applied one level down, and it means the same class of mistake is impossible in the calendar derivations as in the timestamps.

Closed: 512 bits, and what that buys

The tick count is a 512-bit unsigned integer. The ceiling is $2^{512} - 1$, which the system will print for you:

● ● ● *ucal doctor — the domain*

```
domain_max_ticks
13407807929942597099574024998205846127479365820592393377
72356144372176403007354697680187429816690342769003185818
6486050853753882811946569946433649006084095
domain_bits      512
```

In years, that ceiling is about 2.29×10^{103} .

To give that a scale: proton decay, if it happens at all, is expected somewhere around 10^{34} years. The last black holes evaporate around 10^{100} . The domain outlasts both, with room left.

The present epoch sits at about 8.07×10^{60} ticks — which is 6.0×10^{-94} of the range. The counter is, for all practical purposes, still at zero.

A NUMBER THIS BOOK HAD TO CORRECT

RFC UCAL-A1 attaches the figure 7×10^{-17} to this paragraph. That number is real but belongs to a different range: it is the present epoch as a fraction of the **UCID's** 256-bit space, not of the 512-bit domain. The domain fraction is 6.0×10^{-94} .

Rule S is why this correction is here rather than the original figure. The article RFC describes what the book intended to say; the source tree is what is true. This is the first place in the book where those two disagreed, and it will not be the last.

INTERPRETATION

The width was not chosen to be impressive. It was chosen so that **the width never has to change**, because under Rule B the value width is a wire-format commitment: the canonical binary encoding is 64 bytes because the domain is 512 bits, and widening the domain later would invalidate every stored timestamp in existence.

A system that might need to grow its integer has a migration in its future. This one bought its way out of that with sixty-four bytes, once.

Checked: overflow is a typed error

No operation on a time type wraps, and none saturates.

Every arithmetic method that could exceed the domain returns a **Result**. Overflow is **UCAL-E0060**, an ordinary error you handle, not a silently wrong answer you ship. The release profile sets **overflow-checks = true**, so the guarantee

holds in optimised builds and not only in debug ones — which is where this class of bug normally hides.

There is also a workspace lint that fails the build on `wrapping_*` or `saturating_*` appearing on a time type. That is belt and braces, and it is there because the failure mode it guards is the quietest one available: a wrapped tick count is not an error message, it is a timestamp from the wrong end of eternity that looks entirely plausible.

Two backends, one domain

The tick integer has two implementations. The default is a fixed 512-bit stack integer; the alternative is an arbitrary-precision heap integer. They are selected by feature flag and are mutually exclusive at compile time.

The interesting requirement is that they must accept and reject **exactly the same values**. The arbitrary-precision backend is unbounded and would happily represent 2^{513} ; it is made to enforce the same ceiling anyway, and the whole test suite runs against both.

INTERPRETATION

It would have been easier to let the unbounded backend be unbounded. The reason not to is that the two would then be different systems wearing the same name — a timestamp accepted by one and rejected by the other, with no way to tell from the type which you were holding.

The rule that forbids this is one line long, and enforcing it cost a full second copy of the test suite in CI. That ratio — a cheap sentence, an expensive guarantee — is characteristic of the whole design.

No floating point. Anywhere.

This is the constraint with the largest blast radius, so it gets stated in its strongest form: there is no floating-point value in any shipped crate of this workspace. Not in a signature, not in a field, not in a constant, not in an intermediate, and not in the rendering path that produces the human-readable output.

A lint scans the source and fails the build on any float token. It permits exactly one exception — a float reference implementation used as a **test oracle** — and it requires the exception to be marked, and it prints every exemption it honoured on every run, so an escape hatch cannot quietly become a retreat.

What it costs

The cost is real and it is concentrated in one place: cosmology.

Computing the age of the universe at a given redshift means evaluating an integral. In any normal library that is a call to a quadrature routine over `f64`, and it returns a number with an error estimate. Here there is no `f64`, so it is certified interval quadrature over exact rationals — thousands of panels, each bounded above and below, with directed integer square roots underneath.

It is slower by orders of magnitude. The default subdivision depth takes about half a second where a float routine would take microseconds.

What it buys

An **enclosure** rather than an estimate.

The float routine returns a number and a tolerance, and the tolerance is a well-informed guess about accumulated error. The interval routine returns two numbers and a proof that the true value lies between them. Not probably. Provably, given the model.

INTERPRETATION

For most software the float answer is the better engineering, and it is not close. A guess accurate to twelve digits, computed instantly, beats a proof accurate to four, computed slowly.

This project takes the other option because its subject is a quantity whose uncertainty is the interesting part. Part IV is four chapters about an origin that cannot be measured; a cosmology module that reported a confident point estimate would be undermining the book's central claim in the one place where the claim is most testable.

The two widths

One consequence deserves its own note, because it is where the arithmetic discipline becomes an argument.

Every cosmological result carries **two** uncertainty widths, and they are never merged: the width contributed by the quadrature, and the width contributed by the model's own measured parameters. Asking for a single combined tolerance requires calling a method with a name that says you are combining them.

At recombination the quadrature width is about 251 years and the parameter width is about 10,900 years. Merged into one number, the answer would be “about eleven thousand years of uncertainty” — and the fact that forty-three forty-fourths of it comes from the **measurement** rather than from the computation would be invisible.

That distinction is what tells you more computing power would buy nothing here. A merged tolerance hides it. Keeping them apart is a line in the specification, and it is the reason the specification has that line.

WHAT THIS CHAPTER ESTABLISHED

- The domain is unsigned: a result before the datum is `UCAL-E0020`, not a negative number. The same rule applies one level down, to `Ratio`.
- 512 bits reaches 2.29×10^{103} years, past proton decay and black-hole evaporation. The present epoch is 6.0×10^{-94} of it — and the width never has to change, because it is a wire-format commitment.
- Overflow is a typed error, enforced in release builds and by a lint, because a wrapped tick count looks entirely plausible.
- Two backends, one domain: the unbounded one enforces the same ceiling, verified by running the whole suite twice.
- No floating point anywhere. It costs orders of magnitude in the cosmology and buys a certified enclosure instead of an estimated tolerance.

CHAPTER 6

6

Notation

One value, four ways of writing it down. This chapter is about why that is not redundancy, and about what each form is for.

● ● ● *ucal explain — one instant, four notations*

```

ticks
8070205189128471254993117657693008777530466139316558837890625

human      UC1 0031·0687·2481·3000·1638·3018·0779·2671·2006·1837·...

digit5     UC1/5 00000.00000. ... .00111.10222.34411.44000.23023. ...

ucid       0000000000050PM6K45MKCAVY5MPYAMHCJQ142JHE26A2ZAJ9FJ1

```

Two text forms, and why both

The **human form** writes each tier group as a decimal number. The **digit form** writes the underlying base-5 digits directly, five to a group.

They encode the same integer. The difference is who is reading.

A person reading `0031·0687·2481` can see at a glance that this is 31 deeps, 687 drifts, 2481 spans — three quantities in a unit system, each written the way people write numbers. The same prefix in the digit form is `00111.10222.34411`, which is correct, canonical, and unreadable.

INTERPRETATION

The temptation is to call the human form a convenience wrapper and the digit form the real one. That is not quite right, and the distinction matters for Part VIII.

The digit form is canonical **for machines** — parsing, sorting, round-tripping. The human form is canonical **for statements about time**, because tier groups are the units the system actually reasons in. Neither is a rendering of the other; both are renderings of the integer.

Each form carries a tag — `UC1` or `UC1/5` — so a value never travels without declaring which profile and which notation produced it. That is Rule P doing its work at the notation layer: two timestamps from different profiles cannot be compared by accident, because the text itself says they are different things.

Where each form anchors

This is the detail that most repays care, and it is the subject of one of the corrections this project made against its own specification.

A form printed to fewer groups is **less precise**, not **padded with zeros**. Chapter 4 called this “truncation is rounding”. But saying that precisely requires knowing which tier each form counts from.

The human form anchors at **T0**, the beat: the group immediately before the colon. The digit form anchors at **T32**, the top of the ladder. So the same act of dropping trailing groups means the same thing in both — you have said less — but the tier a given group sits at is read from a different reference.

D-A8 — AN AMENDMENT THE IMPLEMENTATION FORCED

The specification originally described the anchoring loosely enough that two readers could disagree about what a truncated form denoted. Implementation made the ambiguity concrete: the human form **cannot express a precision coarser than T0**, because it has no groups above the colon to drop.

That is not a defect. It is a consequence of anchoring at the beat, and it is why the command line falls back to a different form when asked to render a timeline at a tier above T0 — rather than printing a blank field, which is what the first implementation did and which looked exactly like a value.

The canonical binary form

Sixty-four bytes, big-endian, fixed width. Never length-prefixed, never minimal, never a varint.

The width is not negotiable and the reason is worth stating: **byte order is numeric order**. A fixed-width big-endian encoding sorts lexicographically into chronological order, so the encoding is directly usable as a database key or a sort key with no comparator at all.

A minimal encoding — the obvious space optimisation — would break that, and it would break something else: the wire format would depend on the **magnitude**

of the value rather than on the profile. Two timestamps of different sizes would have different lengths, and a reader would have to know the value to know how to read it.

INTERPRETATION

Sixty-four bytes for a timestamp is extravagant by any ordinary standard. Unix time fits in eight.

What the extra fifty-six buy is that the format never has to change. The domain is 512 bits, the encoding is 512 bits, and there is no future in which a stored timestamp becomes unreadable because the range was widened. That is a real cost paid once against a class of migration that software normally pays repeatedly.

UCID: the transcribable form

The fourth notation is a 52-character identifier in Crockford base-32.

• • • *the alphabet*

0123456789ABCDEFGHIJKLMNPQRSTUVWXYZ

I, **L** and **O** are absent because they are confusable with **1** and **0**. **U** is absent to avoid accidental obscenity. The alphabet is strictly ascending in ASCII — and there is a test that checks that rather than trusting it — which is what makes lexicographic order equal numeric order here too.

Why 256 bits and not 512

Because the UCID is meant to be read, typed, and quoted by a person, and the domain is not.

At five bits per character, covering the full 512-bit domain would take **103 characters**. At 256 bits it takes 52, which is 260 bits of capacity — so the leading character encodes a single significant bit and is always **0** or **1**.

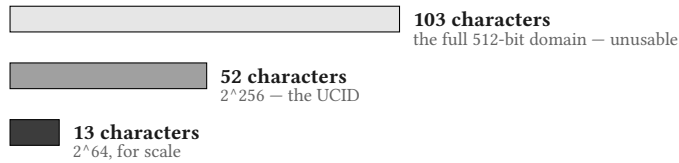


Figure 3 — UCID length against range. Five bits per character, so width is linear in bits — and the domain is twice the bits.

The ceiling that buys is 2^{256} ticks, about 1.98×10^{26} years. That is past the end of the stelliferous era and thirty orders of magnitude beyond the present epoch — and far short of the domain’s 2.29×10^{103} years.

Above the ceiling you do not get a truncated identifier. You get `UCAL-E0031`, **instant outside UCID range**. The same refusal-rather-than-approximation that runs through the whole design.

What UCID is not

The source documentation is emphatic here, and so is this book.

UCID contains no randomness. It is a pure function of the instant. Two events at the same tick receive the same identifier, which makes it useless as a unique key for concurrent events.

It is worse than merely non-random. Chapter 7 shows that an instant read from a nanosecond clock has at least twenty-one trailing base-5 zeros by construction, so the low-order characters are not just predictable but **structurally constrained** — consecutive nanoseconds share more than twenty leading characters.

There is a test called `ucid_has_no_entropy` whose entire job is to measure how badly the identifier would fail if someone used it as a UUID.

INTERPRETATION

Writing a test that demonstrates your own type is unsuitable for an obvious misuse is an unusual thing to do. It costs nothing to omit; nobody would notice.

It is here because the failure would be silent. A UCID **looks** like a UUID — same length class, same character set, same shape in a log line — and the moment someone reaches for it as one, the collisions begin and never announce themselves. A test that fails loudly at the boundary is cheaper than an incident.

Four forms, one integer

To close the chapter, the whole picture in one table.

form	for	width	sorts?
<code>ticks</code>	the value itself, in decimal	variable	no
human	statements about time, by people	variable	only if padded
<code>digit5</code>	parsing, round-tripping, canonical text	variable	only if padded
binary	storage, wire, database keys	64 bytes	yes
UCID	quoting, transcribing, URLs	52 chars	yes

The two that sort are the two that are fixed-width. That is not a coincidence, it is the whole mechanism, and it is why the text forms carry the caveat “only if padded” rather than being quietly assumed to sort.

WHAT THIS CHAPTER ESTABLISHED

- Two text forms encode one integer: the human form in decimal tier groups, the digit form in raw base-5. Neither is a rendering of the other.
- Each form declares its anchor — human at T0, digit at T32 — which is what makes truncation mean **less precise** rather than **padded**.
- Canonical binary is 64 bytes, fixed-width, big-endian, so byte order is chronological order and the format never has to change.
- UCID is 52 characters over 2^{256} because 512 bits would need 103. Beyond the ceiling is `UCAL - E0031`, not a truncation.
- UCID carries no entropy and is not a UUID. A test measures exactly how badly it would fail as one, because that failure would otherwise be silent.

CHAPTER 7

7

The bridge

Chapter 1 said the design's whole question was whether Earth enters **the arithmetic** or at **a declared boundary you can point to**. This chapter is the boundary.

One constant

Earth enters the system through exactly one number.

• • • *the bridge constant*

```
SECOND = 18 548 584 399 861 000 000 000 000 000 000 000 000 000
000 ticks
```

That is how many ticks are in one SI second. It is an exact integer — declared, not computed — and it is the only place in the workspace where a foreign unit is named at all. A lint fails the build if any identifier in the core crate mentions a foreign unit outside the bridge declaration.

INTERPRETATION

The number of things this constant is **not** doing is the point.

It is not a conversion factor applied throughout the code. It is not a scale on the tick. It does not appear in any arithmetic that produces a timestamp from another timestamp. Every operation the system performs on absolute time is integer addition, subtraction and comparison on tick counts, and none of them has ever heard of a second.

`SECOND` is consulted when a foreign value arrives and when one leaves. In between, it is inert.

Why the direction matters

Converting **into** absolute time is multiplication by an integer. Multiplication of integers is exact. So a whole number of seconds becomes a whole number of ticks with no rounding whatsoever, ever, under any circumstances.

Converting **out** is division, which is where rounding lives — and so the rendering path is the only place a rounding mode is chosen, and it is always chosen explicitly by the caller.

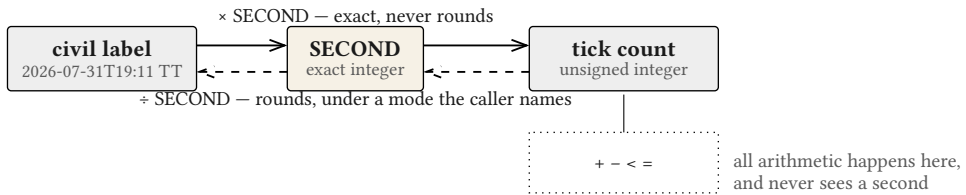


Figure 9 — The bridge. Earth enters and leaves at one declared constant; the arithmetic in between is integer operations on tick counts.

Refusing rather than rounding

Ask the system to accept a time finer than the bridge constant can represent and it does not round. It returns **UCAL-E0043** — **foreign-unit input finer than the bridge constant permits**.

The threshold is 10^{-30} seconds, and it is not arbitrary: it is where the declared constant's own decimal precision runs out. Below it, the system does not know the answer, and it says so instead of inventing one.

THE PATTERN, AGAIN

This is the third time in three chapters. Before the datum: refuse the question (UCAL-E0020). Beyond the UCID range: refuse to truncate (UCAL-E0031). Finer than the bridge: refuse to round (UCAL-E0043).

A system that answered all three would be more convenient and would be lying in three different ways.

The invariants that fall out free

Here is the part that makes the design feel less like a set of choices and more like a consequence.

SECOND was set to a multiple of 10^{30} . Because $10^{30} = 2^{30} \cdot 5^{30}$, that means **SECOND** carries thirty factors of five. The nanosecond — a thousandth of a millionth of it — carries twenty-one.

So:

quantity	factors of 5	consequence
SECOND	30	a whole second lands with 30 trailing base-5 zeros
NANOSECOND	21	a whole nanosecond lands with 21 trailing base-5 zeros
ORIGIN_OFFSET	61	the bridge epoch is zero in every tier below T0

Read that in the notation of chapter 6 and it means something concrete: **any instant that came from a wall clock has a long tail of zeros in its digit form**. Six full tier groups of them, for a nanosecond clock.

INTERPRETATION

These are excellent invariant tests, and the project uses them as such — they are cheap to check and they fail loudly if the bridge is ever mis-declared.

They are also the reason chapter 6 could say the UCID is not merely non-random but **structurally constrained**. The zeros are not a coincidence about particular timestamps; they are a theorem about every timestamp that entered through the bridge. An identifier whose low-order digits are provably zero is not a candidate for a unique key, and it is better to know that from the invariant than to discover it from a collision.

One pivot, and where the leap seconds live

Civil time is a mess, and the design's response is to confine the mess.

There is exactly one pivot scale: **Terrestrial Time**. TT is a uniform scale with no leap seconds, which is precisely why it was chosen — a timestamp converted through TT never has to know about the irregular ones.

Leap seconds exist, and the system knows all twenty-seven of them. They are applied at the parse and format boundary and nowhere else. No arithmetic on absolute time ever encounters one.

• • • *ucal doctor — the leap table*

```

leap_seconds:
  table_version      IERS Bulletin C 70 (no leap second to
2026-06-30)
  entries            27
  complete_through   2026-06-30
  pre_1972           the 1961-1972 rubber-second era is modelled
exactly;
                                UTC before 1961-01-01 is UCAL-E0041
  network            never; the table is bundled and offline

```

Two details in that output are worth pausing on.

The table is bundled and offline. No command in this system performs network access, ever. A time library that phones home is a time library whose answers depend on whether the network was up.

The rubber-second era is modelled exactly. Between 1961 and 1972, UTC did not tick at the same rate as TAI — it ran at declared fractional offsets that changed periodically, and the offsets were not whole seconds. Most software declines to handle this and starts its UTC support in 1972.

INTERPRETATION

It turns out to be exactly representable in this system, and the reason is a small arithmetic accident worth recording. The era's rate coefficients — 0.001296, 0.0011232, 0.002592 seconds per day — are all divisible by 27. A day is 86,400 seconds, which carries $3^3 = 27$. The 27s cancel, and what is left is exact in the rationals the system already uses.

Nobody designed that. It was discovered while implementing, and it meant an era that could have been declared out of scope became eleven years of exactly convertible history instead. Before 1961 there is nothing to convert exactly, and the system returns `UCAL-E0041` rather than guessing.

What the bridge costs

Honesty, since Part III is about what implementation refused.

The bridge is where the claim “no Earth content” is at its weakest. **SECOND** is an Earth-derived constant sitting in the core crate, and the tick’s **length** is fixed by convention against it, as chapter 2 admitted.

What the design achieves is not the elimination of that dependency — that is not available to anyone — but its **localisation**. There is one constant, in one declared place, named by one identifier, with a lint that fails the build if a second one appears. The arithmetic is clean; the boundary is dirty; and the boundary is visible.

INTERPRETATION

Whether that is enough is a fair question and this book does not think it is settled. What can be said is that the alternative designs are worse in a specific way: a system with conversion factors distributed through its operations has the same dependency and no place to point at.

A dependency you can point at can be argued about. One that has been dissolved into the arithmetic cannot.

WHAT THIS CHAPTER ESTABLISHED

- Earth enters through one exact integer constant, **SECOND**, and a lint fails the build if the core names a foreign unit anywhere else.
- Conversion **in** is multiplication and never rounds; conversion **out** is division and is the only place a rounding mode is chosen.
- Input finer than 10^{-30} s is refused, not rounded — the third refusal in three chapters.
- Whole seconds land with 30 trailing base-5 zeros and whole nanoseconds with 21, which makes good invariant tests and explains why UCID has no entropy.
- TT is the only pivot; all 27 leap seconds live at the parse/format boundary, offline. The 1961–1972 rubber-second era turns out exactly representable, by accident.
- The bridge does not eliminate the Earth dependency. It localises it to one place you can point at and argue about.

CHAPTER 8

8

Derived calendars

The ladder of chapter 4 has no days and no years in it, deliberately. But people live on planets, and a calendar that cannot say what day it is locally is an instrument of limited interest.

So there is a second layer: local calendars, derived from a body's own motion. This chapter is how that derivation works. Part V is how far it generalises and where it fails.

Four components

Every calendar in the system is the same shape:

component	what it is	source
Body	rotation, solar day, orbital period — as exact rationals of ticks	cited measurement
Anchor	where the count starts in phase	declared, cited, empirical
LeapRule	intercalation: how the fractional day is absorbed	derived
Cycles	grouping periods, if the calendar names a satellite	derived

Three of the four are computed. One is not, and the one that is not is the interesting one — it gets flagged here and gets a full treatment in chapter 16.

Units, as rationals of ticks

A body’s periods enter as exact rational numbers of ticks, each carrying its citation, its epoch, its secular rate, and the window over which it is valid.

That last part matters more than it sounds. Earth’s rotation is slowing by roughly 1.8 milliseconds per century; its tropical year is shortening by about half a second per century. A parameter treated as a constant is wrong the moment you leave the epoch it was measured at — and this system is aimed at deep time, where “leaving the epoch” is the normal case.

So evaluating a parameter outside its stated window produces a warning rather than a confident extrapolation. The system will tell you the number is out of its depth.

Intercalation is derived, never declared

Here is the heart of the mechanism.

A body's year is not a whole number of its days. Earth takes about 365.2422 rotations per orbit. Some scheme has to absorb the 0.2422, and every civilisation that has built a calendar has invented one.

This system does not invent one. It **computes** the schemes, by expanding the fractional part as a continued fraction and reading off the convergents — the sequence of best rational approximations, in the precise sense that no fraction with a smaller denominator comes closer.

● ● ● *ucal cal show earth-d — the derivation*

```
intercalation:
  whole_days_per_year  365
  rule                 31/128
  bound                1 local day per 10000 local years
  walked:
    1: 1/4             - 1 day slips in 128 local years
    2: 7/29            - 1 day slips in 1234 local years
    3: 8/33            - 1 day slips in 4269 local years
    4: 31/128          - 1 day slips in 400000 local years  <- chosen
    5: 752/3105        - 1 day slips in 62100000 local years
    6: 4543/18758      - 1 day slips in 937900000 local years
```

Read the first line of that walk. Convergent 1 is $1/4$ — one leap day every four years. That is the Julian calendar, derived from Earth's rotation and orbit with no knowledge of Rome.

The system then walks the ladder until a convergent meets the requested drift bound, and reports the entire walk rather than only the winner — so you can see what was rejected and why.

INTERPRETATION

The reporting-the-whole-walk decision looks like a nicety and is not. A mechanism that returns only its answer cannot be argued with. One that shows its working can be checked, and — as chapter 10 will describe at length — can turn out to contradict the person who built it.

The single most important output this project ever produced came from reading a walk and noticing what was **not** in it.

Cycles, and the absence of them

Months are grouping periods, and they come from a satellite.

If a calendar names a grouping satellite, the system derives the cycle structure the same way — continued-fraction expansion of the ratio between the orbital period and the satellite’s synodic period. Earth names the Moon, and the derivation produces 235/19, the Metonic cycle, known since antiquity and recovered here from Earth’s periods alone.

If a calendar names no grouping satellite, it has no months. Not a default month, not a synthesised one, not a fallback to Earth’s. None.

• • • *ucal cal list — three derived calendars*

```
earth-d:
  leap_rule      31/128 (convergent 4)
  cycles         from moon

mars-d:
  leap_rule      45/76 (convergent 6)
  cycles         none — the calendar names no grouping satellite

titan-d:
  status no anchor: complete in units, intercalation and cycles,
          incomplete in phase. Asking for local fields is UCAL-
E0062.
```

MARS HAS NO MONTH, AND THAT IS THE CORRECT OUTPUT

Mars has two satellites. Phobos orbits in about 0.45 of a Martian day; Deimos in about 5.4. Neither is anything a person would call a month.

The system could have synthesised one — divided the Martian year into twelve, say, or borrowed a period from somewhere. It returns nothing instead, because a month-shaped structure imposed on a body that has no month is Earth structure leaking through a mechanism built specifically to keep it out.

The absence is not a gap in the implementation. It is the implementation working.

The anchor: the one thing that cannot be derived

Everything above came out of arithmetic on cited parameters. The anchor does not, and cannot.

Intercalation tells you how long a year is. It does not tell you **when the year starts** — and no amount of tick counting will, because phase is not a consequence of period. You can know exactly how long Earth takes to rotate and still not know whether it is currently noon.

So the anchor is declared: one cited, interval-valued constant per body, with a revision number and an uncertainty window.

A calendar with no anchor is not an error to construct. It is an error to **ask local fields of:** `UCAL-E0062`. Titan is in exactly that state, and chapter 16 explains why it will stay there.

The anchor is where this mechanism admits it is not self-sufficient, and the admission is structural rather than apologetic. It is one declared constant, no more privileged than the rotation period sitting beside it and no less necessary.

Derived and legacy, held apart

52

● ● ● *ucal cal list — the legacy entries*

```

earth-civil:
  kind      legacy – declared tables (§8.6)
  arbitrary 4
  leap_rule 97/400 (NOT a convergent – declared, not derived)

earth-julian:
  kind      legacy – declared tables (§8.6)
  arbitrary 4

```

earth-civil has four items of **arbitrary content** — irregular month lengths, the seven-day week (which corresponds to no astronomical period at all), the leap rule, and the epoch. None of them follows from Earth’s motion. They are declared tables, and the type system keeps them in a separate trait: a function that wants a derived calendar will not accept a legacy one, and there is no blanket conversion.

Note the parenthesis on the leap rule. **97/400** is marked **NOT a convergent**. That label is the subject of chapter 10, and it is where this book’s most uncomfortable finding lives.

INTERPRETATION

It is worth being explicit that “legacy” here is a technical classification and not a judgement. A declared table is not a worse calendar than a derived one; the Gregorian reform solved a real problem and solved it well enough to still be in use.

What the classification records is **where the authority comes from** — arithmetic on cited parameters, or a decision someone made. Part VI applies that same distinction to scriptural chronologies, and it has to apply it evenhandedly or not at all. A system that calls the Gregorian derived and Seder Olam declared would be doing something other than classifying.

WHAT THIS CHAPTER ESTABLISHED

- Every calendar is (Body, Anchor, LeapRule, Cycles). Three are computed; the anchor is not.
- Body parameters are exact rationals of ticks carrying epoch, secular rate, and validity window — and evaluating outside the window warns rather than extrapolating.
- Intercalation is derived by continued-fraction expansion, and the whole walk is reported, not just the winner. Convergent 1 for Earth is $1/4$ — the Julian rule, derived.
- Cycles come from a named grouping satellite or do not exist. Mars has no month, and the absence is the mechanism working.
- The anchor cannot be derived, because phase is not a consequence of period. It is one declared, cited, interval-valued constant, and its absence is `UCAL-E0062` rather than a default.
- Legacy calendars are held apart by the type system. `97/400` is marked **not a convergent** — the subject of chapter 10.

PART III

What implementation refused

Design as something that loses arguments.

CHAPTER 9

9

Divergence

Parts I and II described a system. This part describes the arguments that system won against the person who specified it.

That framing is deliberate. A specification written before implementation is a set of predictions about what will turn out to be buildable, and some of them are wrong. The usual practice is to quietly fix the specification afterwards so that it appears to have been right all along. This chapter is the record of not doing that.

What was checked, and how

Before a line of the library was written, every constant and derivation in the specification was recomputed independently — twice, by two separate integer routes that share no code, and compared against what the document claimed.

That harness still runs:

```
● ● ● cargo run -p xtask
```

```
UC-P0 constants harness – RFC UCAL-1, profile UC-1
```

```
ok    routes A and B agree on every field
ok    §3.3 profile constants are const-constructible and correct
ok    BEAT: 5^60 (computed) == literal
ok    provenance: AGE_s
...
96 passed, 0 failed
```

It produced fifteen findings. Fourteen stand; one was withdrawn, and the withdrawal is the most instructive of the lot.

The map

Twenty-four named rules. Six were amended by contact with the implementation.

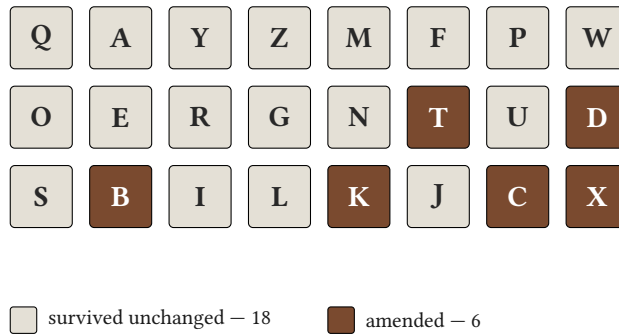


Figure 10 — The twenty-four rules by fate. None was dropped; six changed, and four of the six changed because the implementation proved the original could not be built as written.

rule	delta	what changed
T — truncation is uncertainty	D-A8	what a printed form means , and how each form anchors
D — two text forms	D-A8, D-A4	human anchors at T0, digit at T32; the fixtures were truncated
B — canonical binary	D-A7	full-width encode is 45 divmod steps, not 44
K — one derivation mechanism	D-A5	grouping cycles are declared per body, not admitted by a global bound
C — parameter provenance	D-A11, D-A13	obliquity cannot be a RatedParam ; a drift bound is a rate, not a duration
X — certified enclosures	D-A14	the cosmological integral cannot be quadratured as written

Three kinds of losing

The fourteen findings sort into three classes, and the classes matter more than the count.

Editorial — five findings where the specification stated a number or a description inaccurately, with no behavioural consequence. **ORIGIN_OFFSET** has 61 trailing base-5 zeros, not 62. Appendix B’s seconds column was chained from neighbouring rows instead of computed independently, and drifts in the fifth significant figure. These are the cheap ones.

Correction — six findings where the specification was wrong and the implementation could not follow it. The synodic-period formula computed the wrong

quantity. A drift bound was typed as a duration when it is a rate. The cosmological integral, as written, was improper and could not be certified at all.

Amendment — three findings where the specification was coherent, buildable, and adopted differently on purpose after implementation revealed what it cost.

INTERPRETATION

The amendments are the interesting category, because nothing forced them.

D-A5 is the clearest. The specification admitted a satellite as a “grouping satellite” — a source of months — if its synodic period fell within a bracket of 5 to 100 solar days. That is a perfectly implementable rule. It was replaced because the bracket is **calibrated on Earth’s Moon**, which puts an Earth-derived constant inside the one mechanism whose entire purpose is to keep Earth from being the template.

Nothing broke. The tests passed. The rule was changed because implementing it made visible that “month-like” is not a derivable predicate — **month-like** is an Earth predicate — and a bound tuned until Mars gave the expected answer would have been fitting the constant to a conclusion.

What each change cost

Honesty requires the costs alongside the corrections.

D-A5 cost a calendar the ability to have months by default: a body must now **name** a grouping satellite, with a citation for why. Mars names none, so `mars-d` has no months at all. That is the correct output, and it is also less than the specification promised.

D-A13 cost a type. `max_drift: &Delta` became `DriftBound { days, per_years }` in the body’s own local units — which means the bound cannot be stated in seconds, which means a caller who thinks in seconds has to convert first and think about what they are doing.

D-A14 cost the cosmology module its shape. The specification’s integral runs to infinity; certified quadrature needs a compact interval. The substitution $u = 1/(1 + z)$ fixes it, and finding that substitution was a day of work that the specification’s phrasing implied would not be necessary.

The withdrawal

One finding was raised and retracted, and it stays in the record.

The verification claimed that Appendix C's fixture for 15 March 44 BC was one day late. It was not. The checking oracle was wrong: it applied an era adjustment that exists to make **truncating** integer division behave like flooring, on top of a language whose division already floors. The correction was correct for positive years, which is why seven of the eight fixtures passed and only the one negative year failed.

Appendix C is eight for eight.

INTERPRETATION

A retracted claim is more useful in the record than an absent one, and the reason is not modesty.

A verification process that only ever reports findings is indistinguishable from one that manufactures them. The retraction is the evidence that the oracle was itself checked — that when the specification and the harness disagreed, the harness was not automatically believed.

It is the smallest of the fifteen entries and it does the most work.

Did it diverge enough?

The plan for this book set a condition on Part III existing at all: if the implementation turned out to follow the specification closely, this part would collapse into two pages inside Part II, because a book claiming that implementation talks back has to show that it did.

The answer is yes, and here is the count. Fourteen standing findings. Six of twenty-four rules amended. Six places where the specification was simply wrong. One where its central method could not be executed as written. And separately from the rules, four of the project's six gated experiments hit their kill criteria — including one that fired before implementation started, when the integer library the specification named changed its API mid-build.

THE COUNT IS NOT THE ARGUMENT

Fourteen findings out of a document this size is not a scandal, and it is not presented as one. Specifications of this length routinely contain more.

What Part III claims is narrower: that the findings were **recorded** rather than absorbed, that the record distinguishes what was wrong from what was merely changed, and that one entry is a retraction. Those three properties are what make the next chapter's claim believable, and the next chapter is why this part exists.

WHAT THIS CHAPTER ESTABLISHED

- Every constant and derivation was recomputed by two independent routes **before** the library was written; the harness still runs, 96 checks.
- Fifteen findings: five editorial, six corrections, three amendments — and one withdrawal.
- Six of twenty-four rules were amended. None was dropped.
- The amendments were not forced. D-A5 replaced a working rule because implementing it revealed an Earth constant inside the mechanism built to exclude Earth.
- Each change cost something, and the costs are recorded next to the corrections.
- The withdrawal stays in the record, because a process that only ever finds things is indistinguishable from one that invents them.

CHAPTER 10

10

The 97/400 correction

This is the chapter the book exists for.

Everything else here is a claim that a program can do philosophical work. This chapter is the evidence, and the evidence takes the only form that counts: the program produced a result its author did not want, about a claim its author had published twice, and the author printed it.

The claim that was made

Chapter 8 described the intercalation mechanism. A body's year is not a whole number of its days; the fraction is expanded as a continued fraction; the convergents are the best rational approximations, in the precise sense that no fraction with a smaller denominator comes closer.

The specification, in two successive revisions, said this about it:

- | The mechanism reproduces the Julian and Gregorian rules as convergents.

It is a satisfying sentence. It says the machinery, given nothing but Earth's rotation and orbit, rediscovers the two calendars Western civilisation actually built. If true, it would be the strongest possible demonstration that the derivation is not merely consistent but **correct** — that it finds what people found, from first principles.

Half of it is true.

What the mechanism actually returns

● ● ● *ucal cal show earth-d — the full walk*

```
intercalation:
  whole_days_per_year  365
  rule                 31/128
  bound                1 local day per 10000 local years
  walked:
    1: 1/4             - 1 day slips in 128 local years
    2: 7/29            - 1 day slips in 1234 local years
    3: 8/33            - 1 day slips in 4269 local years
    4: 31/128          - 1 day slips in 400000 local years  <- chosen
    5: 752/3105        - 1 day slips in 62100000 local years
    6: 4543/18758      - 1 day slips in 937900000 local years
    7: 9838/40621      - 1 day slips in 4062100000 local years
    8: 24219/100000    - 1 day slips in never (exact) local years
```

Convergent 1 is $1/4$. One leap day every four years. That is the Julian calendar, derived from Earth's motion with no knowledge of Rome, and the first half of the claim is confirmed exactly.

Now look for $97/400$.

It is not there. It is not at depth 4, where the chosen rule sits. It is not at depth 8, where the expansion terminates because the parameter is exact. **It is not at any depth**, and there is a check in the constants harness whose only job is to assert its absence.

Why it cannot be there

This is not an accident of where the walk stopped. It is a theorem.

The convergents of a continued fraction have a defining property: convergent p/q is the **best** rational approximation with denominator at most q . Nothing with a smaller or equal denominator gets closer. So if $97/400$ were a convergent, no fraction with a denominator below 400 could beat it.

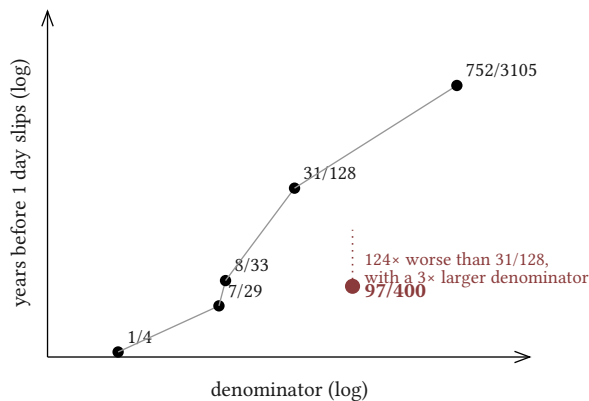


Figure 6 — Earth's convergent ladder, and the Gregorian rule off it. Every point on the line is the best approximation available at its denominator. $97/400$ is not on the line.

Two fractions with smaller denominators beat it:

rule	denominator	1 day slips in	against $97/400$
8/33	33	4,269 yr	1.32× more accurate, denominator 12.1× smaller
31/128	128	400,000 yr	124× more accurate , denominator 3.1× smaller

97/400

400

3,226 yr — the Gregorian rule

31/128 is a hundred and twenty-four times more accurate than the Gregorian rule while being a simpler fraction. That is the finding, and it is not close.

What that means, and what it does not

It does not mean the Gregorian calendar is bad. The reform of 1582 solved the problem it was aimed at — Easter drifting against the equinox — and it solved it well enough to still be in use four and a half centuries later. Accuracy against the tropical year was one constraint among several, and it was not the binding one.

It does not mean 97/400 was a mistake. A rule of 400 years is expressible as “every four years, except centuries, except every fourth century,” which a person can apply without arithmetic. **31/128** cannot be stated that way at all.

What it means is precisely this: **the Gregorian rule is a declared table, not a derivation.** Chapter 8’s classification — legacy versus derived — is not a value judgement dressed up as taxonomy. It is a factual claim about where a rule’s authority comes from, and here is the case where it bites.

INTERPRETATION

The interesting thing is not that the specification was wrong about a fraction. It is that the specification was wrong **in the direction of its own thesis.**

A machinery that rediscovers both historical calendars is a better story than one that rediscovers one and quietly demonstrates the other was never derived. The sentence was not a careless error; it was the error the argument wanted to be true. It survived two revisions because nobody checked the half that was flattering.

Maimonides' charge

There is a nine-hundred-year-old objection to systems like this one, and it is worth stating in its own words because the correction above is the only kind of answer it admits.

In the **Guide of the Perplexed** I.73, Maimonides considers the Kalām practice of constructing metaphysical premises and observes — about people he otherwise treats with respect — that their premises were not derived from the world but assembled to make a desired conclusion provable. They knew where they were going and built the road backwards from it.

The charge generalises immediately to any formal system built by someone with a thesis. A calendar that derives calendars, built by a person who believes derivation is superior to declaration, is exactly the sort of thing that could be tuned until it produced the flattering result. The drift bound could be adjusted. The parameter could be chosen. The depth could be stopped where the answer looked good.

INTERPRETATION

You cannot answer that charge with assurance, because assurance is what it is about. You can only answer it with evidence, and the evidence has to be a case where the system contradicted its builder on something he had already committed to in public.

This is that case. The claim was published twice. The mechanism said no. The mechanism’s answer was recorded, the two revisions that said otherwise are named as wrong, and the harness now contains an assertion whose only purpose is to keep the correction from being quietly reversed.

That is not proof of good faith — nothing is. It is the strongest available substitute: a system that has demonstrably been permitted to answer back.

What was done about it

Four things, and the fourth is the one that matters in five years.

The specification’s sentence was corrected rather than deleted, and the revisions that carried it are named.

The mechanism was left alone. No bound was adjusted, no parameter retuned, no depth extended in the hope that 97/400 would appear further along. It does not appear further along; the expansion terminates at depth 8 because Earth’s parameter is exact in the system, and the fraction is absent from all eight.

`earth-civil` was reclassified as a legacy calendar, and its leap rule carries the label in the tooling's own output:

• • • *ucal cal list*

```
earth-civil:
  kind      legacy – declared tables (§8.6)
  arbitrary 4
  leap_rule 97/400 (NOT a convergent – declared, not derived)
```

And a check was added to the constants harness asserting that `97/400` is absent from Earth's convergents at every depth. It runs on every build. If someone ever changes a parameter in a way that makes the flattering claim true, the build will say so — which is the only form of vigilance that survives the author losing interest.

WHAT THIS CHAPTER ESTABLISHED

- The specification claimed twice that the mechanism reproduces the Julian **and** Gregorian rules as convergents.
- The Julian rule is convergent 1, exactly as claimed. `97/400` is not a convergent at any depth, and it cannot be: two simpler fractions beat it.
- `31/128` is 124× more accurate with a denominator 3.1× smaller; `8/33` beats it with a denominator 12.1× smaller.
- This is not a criticism of the Gregorian reform. It is the factual content of the legacy/derived distinction — where a rule's authority comes from.
- The error ran in the direction of the author's own thesis, which is why it survived two revisions.
- Maimonides' charge in **Guide** I.73 admits only one kind of answer: a case where the system contradicted its builder in public. This is that case, and a build-time assertion now keeps the correction from being reversed.

PART IV

The datum

Why it cannot be measured, and what was done instead.

CHAPTER 11

11

Why it cannot be measured

Chapter 3 gave one reason and promised three more. Here they are, and then a fourth the specification does not state, which is the best of them.

The question this part answers is narrow and it is worth keeping in view: **why is the origin of an exact integer count not simply the measured age of the universe?** The measurement exists. It is very good. Why not use it?

First: exactness cannot come from measurement

The published age is 13.787 billion years, plus or minus 0.020 billion.

The plus-or-minus is not a formality. Converted into ticks it is a number with fifty-eight digits — about 0.145% of the entire span from the datum to now, or 141.53 drifts on the ladder of chapter 4. If tick zero were **defined** as the measured age, every timestamp would be an offset from a foundation that wobbles by that much, and the exactness of the integer arithmetic would be decoration over a guess.

This is the reason that forces the decision. The others explain why the decision is not a workaround.

Second: the limit is not an event

The FLRW $t \rightarrow 0$ limit is not an observable event. It is where a mathematical model's coordinate goes when you extrapolate backwards, and the model is known to stop describing anything real well before you get there.

You cannot measure the time of an event that the theory containing it declines to assert exists. What is measured is the **age** — how long the universe has been expanding under the model — which is a different quantity that happens to be numerically what you want.

Third: the extrapolation is model-dependent

The figure 13.787 is not read off an instrument. It is inferred: a set of parameters is fitted to observations, and the age falls out of the fit. Change the model and the number changes.

A datum defined by that inference would inherit not only its error bar but its **dependence on a cosmological model that will be revised**. Timestamps written today would mean something slightly different after the next data release — which is failure mode F1, and the reason profiles exist.

Fourth: Kant, and the question's shape

The specification stops at three. There is a fourth, older than all of them, and it is the one that makes the other three feel less like obstacles and more like symptoms.

In the **Critique of Pure Reason**, the First Antinomy sets out a thesis — that the world has a beginning in time — and an antithesis — that it does not — and argues that **both proofs are valid**.

Kant's resolution is that both conclusions are false, because both share a premise that cannot be granted: that the world-series is a completed totality, given as a whole, about which a question of finite-or-infinite extent has an answer waiting to be found. On his account the series of past states is not an object of possible experience at all; it is something reason extends indefinitely, never something it is handed.

If that is right, then the search for the true origin is not a hard empirical problem awaiting better instruments. It is a well-formed-looking question with nothing of the appropriate kind to be true **of**.

INTERPRETATION

This is the strongest available support for stipulating the datum, and it is worth being precise about what it does and does not do.

It does not show that the universe has no beginning. Kant is explicit that the antithesis fails too. What it shows is that the **question as posed** — where is the real zero, so that we may count from it — is asking for something that could not be supplied even in principle.

Which means stipulating is not a retreat from a better method. There is no better method. Chapter 3 said the system refuses the question rather than answering it with a negative number; this is the philosophical form of the same refusal, and it arrived two centuries before the measurement did.

The book returns to Kant once more, in its last chapter, for the part of his argument that is less comfortable.

The datum is in ordinary company

All of that can make a stipulated origin sound exotic. It is the normal case.

epoch	zero	chosen because
TAI	1 January 1958	international agreement; atomic clocks were running
Julian Day	1 January 4713 BC	a convenient common multiple of three cycles
Unix	1 January 1970	recent, round, and before the machines that used it
UC-1	the datum	an exact count needs an exact origin

None of the first three is a discovery about the universe. All are decisions that turned out to be useful, and nobody finds them troubling.

THE EXACT PARALLEL: THE SI SECOND

The second is the sharpest case, because it is doing the same thing in the same system.

The SI second is defined as 9,192,631,770 periods of a caesium-133 hyperfine transition. That number is not a fact about caesium that was waiting to be found. It was **chosen**, in 1967, to match the ephemeris second then in use — itself defined from a particular fraction of a particular tropical year.

So the second is a stipulation calibrated against an older stipulation, and the calibration carries an uncertainty that the definition does not inherit. Nobody says the second is uncertain because the 1900 tropical year was imperfectly measured. The definition is exact; the **correspondence** to what it was calibrated against is not.

That is precisely the structure of the datum, and it is why the datum is unremarkable once you see where it already lives.

What is left over

Stipulation solves the exactness problem completely and leaves a different problem entirely intact.

The physical claim is still true or false. Something **is** the case about the relationship between tick zero and the early universe, and the published uncertainty in that relationship is real information that a serious system should not discard.

So there are two things to hold at once: an origin that is exact by definition, and a claim about that origin that is uncertain by measurement. Conflate them and you get either a false precision or a useless one.

Chapter 12 is how they were kept apart — and the answer is the reason this project became a book rather than a library with a good README.

WHAT THIS CHAPTER ESTABLISHED

- The measured age carries ± 0.020 Gyr — 141.53 drifts, 0.145% of the span. An exact count cannot inherit that.
- The FLRW $t \rightarrow 0$ limit is not an observable event; what is measured is the age under a model, which is a different quantity.
- The inference is model-dependent, so a datum defined from it would shift with each revision — failure mode F1.
- Kant’s First Antinomy adds a fourth: the question presupposes a completed totality that is not a possible object of knowledge, so there is nothing of that kind to find.
- Stipulated origins are ordinary — TAI, Julian Day, Unix — and the SI second is the exact parallel: a definition calibrated against something uncertain, without inheriting the uncertainty.
- What remains is a real claim about the world that must be kept, and kept separate.

CHAPTER 12

12

What was done instead

Three moves. Stipulate the datum. Declare the physical claim separately. Make the claim impossible to compute with.

The first two are ordinary good practice. The third is the one this book is about, and it is the only place where a code listing earns its space.

The exhibit

`BIG_BANG_CLAIM` is a value of type `SignedWindow`. Here is the type's entire definition:

```
pub struct SignedWindow {
    lo: Signed,
    hi: Signed,
}
```

Two fields. What matters is everything that is absent, and the source says so explicitly:

This type deliberately has:

- no arithmetic operators of any kind,
- no `From<SignedWindow>` for `Delta`, `Instant` or `Window`,
- no method returning any of those types.

It cannot be added to an `Instant` and cannot be widened into one. Attempting to use it as an operand is `UCAL-E0025`, and the type system is what makes that unreachable rather than a runtime check.

The claim is fully available. You can read its bounds, render it, print its citation, compare it to itself. What you cannot do is **arithmetic** with it. There is no addition, no conversion, no escape hatch — and no runtime guard either, because a runtime guard is a check that can be forgotten, disabled, or bypassed by a code path nobody thought about.

The tests that prove the absence

An absence is hard to test. You cannot write an assertion that a method does not exist, because the file would not compile if you named it.

So the project uses compile-fail tests: programs that are **required to fail to build**, checked on every run.

● ● ● *tests/compile_fail/signed_window_arithmetic.rs*

```
use ucal_core::{Profile, UC1};

fn main() {
    let a = UC1::big_bang_claim();
    let b = UC1::big_bang_claim();
    let _ = a + b;
}
```

● ● ● *tests/compile_fail/signed_window_as_operand.rs*

```
use ucal_core::{Instant, Profile, UC1};

fn main() {
    let t: Instant<UC1> = Instant::zero();
    let claim = UC1::big_bang_claim();
    // A SignedWindow is not a Delta and must never become one.
    let _ = t.checked_add(&claim);
}
```

● ● ● *tests/compile_fail/signed_window_into_delta.rs*

```
use ucal_core::{Delta, Profile, UC1};

fn main() {
    let claim = UC1::big_bang_claim();
    let _d: Delta = claim.into();
}
```

Each of these is a person trying, in the most natural way available, to use the uncertainty in the age of the universe as a number. Each fails to compile. The test suite passes precisely because they do.

INTERPRETATION

This is the whole thesis in nine lines of Rust, and it is worth spelling out why it is different in kind from a comment saying **do not do this**.

A comment addresses a reader who is paying attention. A runtime check addresses a program that reaches a particular line. A type addresses **everyone who ever writes code against this library**, including the author in three years, including someone who has not read a word of this book, including someone who actively disagrees with the distinction being enforced.

They will not be persuaded. They will be refused, by a machine, in under a second, with an error message. That is a different relationship between an argument and its audience than any essay can have.

Why the type is signed at all

A detail that is easy to pass over and is load-bearing.

The window is **signed** — it can express values before the datum — while **Instant** is unsigned and cannot. That is not an inconsistency. It is the reason the type had to be separate.

The FLRW limit may lie before tick zero. The published uncertainty is symmetric: ± 0.020 Gyr around the datum, so its lower bound is negative. A type that could not express that would have forced the claim to be truncated at zero, which would have misreported the measurement.

INTERPRETATION

So the design faced a genuine dilemma. The claim needs a representation the timeline cannot have; the timeline needs a discipline the claim would break.

Splitting them into two types with no bridge is not a compromise between those requirements — it satisfies both exactly. The claim keeps its sign and its full symmetric range. The timeline keeps its unsigned domain. And the absence of any conversion is what stops the first from contaminating the second.

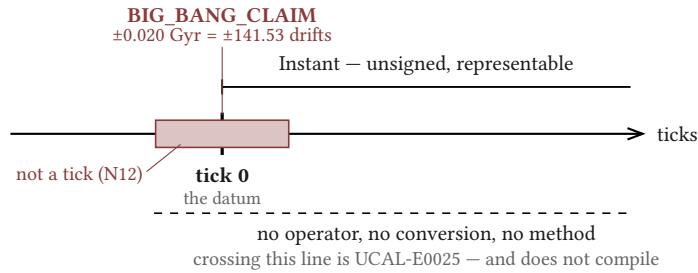


Figure 5 — **BIG_BANG_CLAIM** against the timeline. The claim straddles the datum and extends where no tick exists; the timeline is unsigned. Nothing converts between them.

The provenance chain

The second move — declaring the claim separately — would be worth little if the declaration were prose. It is data, and it re-executes.

• • • *ucal datum — the chain*

```
datum_provenance:
  input      13.787 Gyr +/- 0.020 Gyr (age_of_universe)
  citation   Planck 2018 results VI, A&A 641, A6 (2020)
  unit_defs:
    Gyr = 10^9 x 31 557 600 s (Julian years, exact)
  chain:
    AGE_s      = 13 787 000 000 x 31 557 600
                = 435 084 631 200 000 000 s      (exact)
    AGE_ticks  = AGE_s x SECOND
                = 8070204002895596516263200000... (exact)
    beats      = round_half_even(AGE_ticks / BEAT)
                = 9 304 311 741 502 590 385
    ORIGIN_OFFSET
                = beats x BEAT
  residual_ticks      -3188569143643628194695338...
  residual_rendered   -0.017190364 s
  rationale           a whole-beat datum makes all sub-beat
                      digits of the bridge epoch zero (2.4)
```

Every step is there: the cited input, the unit definitions, each exact multiplication, the rounding to a whole beat, and — the part that matters most — the **residual that the rounding discarded**.

Seventeen milliseconds. The datum is not the published age; it is the published age rounded to a whole beat, and it differs from the input by 0.017190364 seconds. The system prints that number rather than absorbing it.

INTERPRETATION

A provenance chain that reports its own rounding error is doing something a citation cannot. A citation says **where the number came from**. This says where the number came from **and what happened to it on the way in** — which is the part that a reader checking the work would otherwise have to reconstruct.

The chain is also re-executable, which means the claim “this constant follows from that measurement” is testable rather than assertable. The constants harness does exactly that on every run, along two independent integer routes.

Kant policed it by discipline; the crate polices it by the compiler

Chapter 11 borrowed Kant’s First Antinomy. Here is the borrowing that goes deeper.

Kant distinguishes **constitutive** principles, which tell you what objects are, from **regulative** ones, which tell you how to go on investigating. The idea of the world-whole is regulative: it directs reason to keep extending the series and never licenses the assertion that the completed series exists.

Transcendental illusion, for Kant, is precisely the slide from the second to the first — taking a rule for how to proceed as a description of an object.

The datum is a regulative posit. It says **count from here**; it does not say **here is where time began**. The physical claim is the constitutive-looking statement, and it is the one that has been rendered inert.

INTERPRETATION

Kant had no mechanism for this. He had argument, and vigilance, and the expectation that a careful reader would maintain the distinction under pressure. He says himself that the illusion does not go away once diagnosed.

What this project has is not a better argument. It is a worse philosopher with a compiler: the distinction is maintained by something that does not get tired, does not skim, and does not want the flattering answer.

Whether that counts as philosophical work or merely as engineering hygiene is the question Part VIII exists to argue. The claim made here is narrower and, I think, hard to dispute — that a distinction which previously depended on the reader's attention now does not.

UC- Θ , unbuilt

One thing this part must not do is imply the design is finished.

There is a second profile that has been specified and not built. Call it UC- Θ . In it, the datum is not the FLRW limit but the beginning of time **at organization** — a cosmology in which matter is organised rather than created from nothing, so that a physical origin lies later than the reckoning's zero, and there is a "before" that is not a **when**.

INTERPRETATION

Two things make UC- Θ interesting rather than idle.

The unsigned domain, which under UC-1 is a limitation to be defended, becomes **room** — the interval between the datum and the physical origin is representable rather than refused.

And Rule X becomes inapplicable. The cosmology module derives ages under a declared Λ CDM model anchored on the FLRW limit; move the datum off that limit and the certified enclosures no longer mean what they meant. A whole subsystem would have to be re-argued rather than re-parameterised.

That is what makes the profile mechanism more than versioning. Two profiles here are not two configurations of one system. They are two cosmologies, and Part VI, chapter 21 has to say plainly that UC- Θ is heterodox by the standard of the tradition that supplied chapter 13's cautionary tale.

It is listed as unbuilt because it is unbuilt. If it is ever implemented, this section becomes a worked second datum and the book will have to say what changed.

WHAT THIS CHAPTER ESTABLISHED

- `BIG_BANG_CLAIM` is a `SignedWindow`: two fields, and no arithmetic operators, conversions, or methods returning a computable type.
- Three compile-fail tests encode the absence — each is a natural attempt to use the claim as a number, and each must fail to build.
- The type is signed because the claim straddles the datum; `Instant` is unsigned. Splitting them satisfies both requirements exactly rather than compromising between them.
- `datum_provenance` re-executes: cited input, exact chain, and the -0.017190364 s residual that rounding to a whole beat discarded.
- Kant maintained the constitutive/regulative distinction by vigilance and said the illusion persists after diagnosis. Here it is maintained by something that does not get tired.
- UC- Θ is specified and unbuilt. Under it the unsigned domain becomes room, and Rule X becomes inapplicable — which is why profiles are cosmologies, not configurations.

CHAPTER 13

13

Florensky's radius

Chapter 12 described a rule against letting a formal artifact be read as a physical fact. Rules against things exist because the things happen.

This chapter is about the clearest case of it happening, in a book by a mathematician who was better at this than the present author, published in 1922, which contains both the strongest precedent for what this project attempts and the exact failure it guards against — separated by a few pages.

Мнимости в геометрии

Imaginarities in Geometry, by Pavel Florensky: mathematician, electrical engineer, Orthodox priest, and the author of a study of geometry on the projective plane.

The book's project is to give geometric meaning to imaginary quantities, and most of it is exactly what it says. Then, at the end, it turns to Dante.

What Florensky got right

The achievement first, because it is real and because the failure is only interesting if the author was capable.

Consider the **Comedy**'s trajectory. Dante and Virgil descend through the earth to its centre, pass the point of gravity, and continue — and emerge at the Mount of Purgatory on the far side. Dante then ascends through the spheres, past the fixed stars, to the Empyrean, and the poem ends.

In a Euclidean cosmos the trajectory does not close. You go down, keep going, come out somewhere else, go up a long way, and stop. Standard diagrams of Dante's universe handle this by not quite drawing it.

Florensky's observation is that the journey closes — that it returns coherently to its start — if the cosmos is **finite and non-orientable**. The traveller who passes through the centre and continues arrives back, reoriented, without ever turning around.

That is a genuine piece of mathematical reading. It takes a structural feature of a poem seriously enough to ask what geometry the poem presupposes, and the answer is neither trivial nor forced. Whatever one thinks of the rest, this part is good work.

What Florensky did next

In §9 of the same book, he turns to relativity.

The argument runs: consider a cosmos rotating rigidly. At a sufficient radius the co-rotating velocity would exceed c . In the Lorentz factor $\sqrt{1 - v^2/c^2}$, that makes

the quantity under the root negative, and lengths and durations become imaginary. He computes the radius at which this happens. It comes out near the distance to the sphere of the fixed stars in the Ptolemaic scheme.

And then he identifies that surface with the Empyrean.

INTERPRETATION

The distance between the Dante geometry and this is exactly one move, and it is worth naming precisely, because from the outside they look like the same kind of reasoning.

In the first, a formal structure is used to **interpret** a text: the poem's trajectory has a property, and a geometry in which that property holds is exhibited. The claim is about the poem.

In the second, a formal artifact — the imaginary value of an expression outside its domain of validity — is treated as **designating a place**. The claim is about the world, and it is licensed by nothing except that the number came out at a suggestive magnitude.

One move. There is no third thing between them, and no bright line in the prose where the register changes.

What was missing

Florensky had no rule against the second move.

That is not a moral observation. He had a formal apparatus, a serious theological project, and enormous mathematical facility, and no principle in his method that said: **here is where the mathematics stops describing and starts being read**. Nothing in his toolkit distinguished a structure that carries meaning from a structure that is a physical fact.

INTERPRETATION

This project has such a rule, and chapter 12 is what it looks like implemented: a claim about the world, declared with its citation and its magnitude, and made incapable of entering a computation.

It would be comfortable to conclude that the rule makes this project safe from Florensky's error. It does not, and saying so is the point of putting this chapter in Part IV rather than in Part VI.

What the rule protects is the **arithmetic**. **BIG_BANG_CLAIM** cannot become an operand. But nothing prevents a reader — or the author, on a bad day — from looking at a 61-digit integer and reading it as a fact about being rather than a count under a stipulation. That is a failure of interpretation, not of arithmetic, and no type system reaches it.

The last chapter of this book is about exactly that residue.

Basil, fifteen centuries earlier

The insight Florensky's §9 lacked had been available for a very long time.

Basil of Caesarea, in the **Hexaemeron**, addresses what “in the beginning” means and answers with an image: the beginning of a road is not itself the road, and the beginning of a house is not the house. A beginning is not a member of the series it begins.

Applied to time: the beginning of time is not a time. It is not a very short duration, not the first instant of a sequence, not a moment you could in principle point at. It is of a different kind entirely.

That is Rule Q's content, stated in the fourth century with no arithmetic whatsoever. The datum is not the first tick of the universe; it is where counting starts, and those are different sorts of thing.

WHAT THIS CONVERGENCE IS AND IS NOT

Basil is not being credited with anticipating anything, and Rule Q is not being validated by his agreement. Chapter 21 handles the Patristic material properly, including where it conflicts with this project.

The point here is narrower and it is about Florensky. The distinction he needed in §9 was not obscure, not recent, and not outside his tradition — he was an Orthodox priest, and Basil is not a marginal figure in it. He had every resource required to see the problem, and the problem was not visible from where he stood.

That is the useful version of the cautionary tale. Not **he lacked something**, but **he had it and it did not help**, because nothing in his working method forced him to reach for it at the moment it was needed.

What it cost

The chapter should not end on a methodological note, because the people in it did not.

Florensky was persecuted from the early 1920s onward. He continued working — on electrical engineering, on materials science, producing real technical results — under conditions that grew steadily worse. He was arrested, sent to Solovki, and executed in 1937.

Alexei Losev, whose work on number and name appears in chapter 25 and whose position Rule N declines, was imprisoned and sent to the White Sea Canal construction. He went nearly blind there. He survived, and spent decades publishing under constraints that required him to disguise what he was arguing.

INTERPRETATION

It matters that this book's clearest cautionary tale is a man who was killed for the intellectual tradition the tale is drawn from.

The failure identified in §9 is a real failure and this chapter does not soften it. But there is a way of writing about historical mistakes that treats their

authors as material, and it is available here in a particularly cheap form: the mathematician who confused a formal artifact with a place, held up as a lesson.

He was a serious thinker working with the resources he had, on a question this project also finds worth asking, and he paid a price for the asking that no one writing today is being asked to pay. The rule in chapter 12 exists because his error is easy, not because he was careless.

WHAT THIS CHAPTER ESTABLISHED

- Florensky's reading of the **Comedy** — that its trajectory closes only in a finite, non-orientable cosmos — is genuine mathematical work on a real structural feature of the poem.
- In §9 of the same book he computes a radius from the Lorentz factor outside its domain of validity and identifies the resulting surface with the Empyrean.
- The distance between the two is exactly one move: interpreting a structure, versus asserting that a formal artifact designates a place.
- He had no rule against the second move, and Basil's road-and-house image — the distinction he needed — was fifteen centuries old and inside his own tradition.
- Chapter 12's rule protects the arithmetic. It does not protect the reading, and the last chapter of this book is about that residue.
- Florensky was executed in 1937; Losev went nearly blind at Belomorkanal. The error is easy, which is why the rule exists — not because its author was careless.

PART V

Any celestial body

*How far the approach generalises, and precisely where it
fails.*

CHAPTER 14

14

Universal baselines

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Nothing in the arithmetic references a rotation, an orbit, or a civil calendar.

- One derivation mechanism — every calendar is (Body, Anchor, Cycles, LeapRule)
— with Earth as an ordinary instance and no crate named after a body.

CHAPTER 15

15

Deriving a calendar

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Earth: convergents $1/4$, $7/29$, $8/33$, $31/128$ — the Julian rule as convergent 1, the Gregorian absent.

- Mars: 668.59 sols per year, convergent 16/27.
- Titan: tidally locked, handled with no special case — which is the point.
- The Metonic cycle 235/19 derived from Earth's periods unaided.

CHAPTER 16

16

Where it breaks

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 — no shorter than ch. 15 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- The anchor is empirical and cannot be derived.

- A body with no qualifying satellite has no month — Mars is the worked case, and the absence is the correct output.
- A rogue planet has no year.
- A tidally locked body's day equals its orbit.
- Relativistic environments are out of scope.
- Body parameters carry secular rates and validity windows and are wrong outside them — which matters most at exactly the deep-time scale this project targets.

CHAPTER 17

17

The one qualification

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 — one page and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

Specified content

- For a single planet this is a curiosity. For timekeeping across two or more bodies, a universal ladder with local overlays may be the only coherent arrangement.

- A claim about coherence, not a recommendation for adoption.

PART VI

Nine readings

*The traditions as readers of the artifact — neither
validating it nor validated by it. Every chapter names its
conflict.*

CHAPTER 18

18

Greek

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A1 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Archimedes' **Sand Reckoner** builds a positional hierarchy to make a cosmic magnitude expressible, arriving at $\sim 10^{63}$.

- Euclid X.2's anthyphairesis **is** the intercalation algorithm.
- **Conflict:** Aristotle holds time a continuum and the instant a limit; the instrument is discrete. And **Physics** IV.14 makes the datum's objectivity depend on there being a counter.

CHAPTER 19

19

Jewish

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A2 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

Specified content

- **Molad tohu** — “the new moon of chaos” — a stipulated epoch placed before the event it anchors, nineteen centuries early.

- The **helek** chosen for divisibility, exactly as **SECOND** was.
- **Conflict:** the compressed Persian period is provenance overruled by doctrine, and the crate's re-executable chain would expose it — so the instrument judges a tradition, which the book's own rules forbid the author from doing.

CHAPTER 20

20

Islamic

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A3 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

Specified content

- Ghazālī’s ‘āda — divine custom yielding practical certainty without necessity — is what a validity window encodes.

- **Conflict:** under Premise 6 an orbital period is a habit, so a guaranteed drift bound over 400,000 years is meaningless. The crate's ontology is Rushdian and it took a side without arguing for it.

CHAPTER 21

21

Patristic and Latin

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A4 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

Specified content

- Basil’s road-and-house image is Rule Q’s content exactly.
- Augustine’s answer is `UCAL - E0020`: an error, not a negative number.

- **Conflict: *ex nihilo*** is held firmly, so UC- Θ is heterodox by this standard — and the book must say so rather than soften it.

CHAPTER 22

22

Orthodox

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A5 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

Specified content

- Gregory of Nyssa’s διάστημα makes a calendar an instrument of the created order by definition — the limitation is the doctrine, not a defect.

- Palamas' essence/energies distinction as the strongest licence the thesis receives from any tradition.
- **Conflict:** Rule N sides with the 1913 Synodal position against Florensky and Losev — the two thinkers this book most depends on.

CHAPTER 23

23

Latter-day Saint

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A6 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

Specified content

- D&C 130:4–5 states Rule K’s thesis in 1843: reckoning is body-relative.
- Abraham 3:4 is a declared bridge constant with a profile tag on each side.

- **Conflict:** Kolob is a governing **body**, where Rule K.5 permits only an abstract ladder — the privilege is located differently.

CHAPTER 24

24

Modern philosophy

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A7 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- The First Antinomy supplies a fourth reason for Rule Q the specification does not state.

- Constitutive versus regulative, enforced by a type with no arithmetic: Kant policed the boundary by discipline, the crate by the compiler.
- **Conflict:** Kant's number is the schema of magnitude — monadic — which contradicts the eidetic number of A1 and A8, and Rule G asserts both readings of one integer.

CHAPTER 25

25

Russian

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A8 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Fyodorov makes total addressability of the past an obligation.

- Vernadsky's process-relative times and Chizhevsky's solar-period dating as Rule K's unacknowledged ancestry, a century early.
- **Conflict:** Bugaev proposes discreteness **as worldview**, which the book declines — from the closest kin. And Losev's *имя* makes the name constitutive where Rule N makes it a locale-table entry.

CHAPTER 26

26

Method

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §17.2 — A9 and has not been written. Its outline is below so that the book’s shape is visible and its length is not quietly understated.

Specified content

- Klein on **arithmos**; Sorabji on time, creation, and the continuum.

- **Conflict:** the method makes the code the invariant and the traditions the variables. That is itself a metaphysical choice, made without argument — and disclosing it is the last and most self-referential application of Rule M.

PART VII

The instrument as research tool

*What a program can do that an argument cannot: it can
be run against material its author did not choose, and
return an answer he did not want.*

CHAPTER 27

27

Six samples

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §18 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- S1 — comparative chronology on one axis: Seder Olam, Byzantine, Ussher, D&C 77, each a declared profile.

- S2 — calendar audit by convergent. Julian **is** convergent 1; Gregorian is not a convergent at any depth; Jalali 8/33 **is** convergent 3.
- S3 — uncertainty audit: cited versus stipulated.
- S4 — cross-body simultaneity: the demonstration that “now” is not a shared object.
- S5 — measuring διάστημα with no Earth content in the units.
- S6 — a revealed ratio evaluated as a bridge constant, refused as an anchor.

CHAPTER 28

28

Null results

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Each sample's negative finding, stated as plainly as its positive one.

- Where a sample produced nothing, say so; where a result was weaker than hoped, say how much.
- This chapter is what distinguishes the book from a demonstration.

PART VIII

The claim

CHAPTER 29

29

Why a program

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- An essay can assert that a distinction ought to be respected; a type system can make violating it fail to build.

- Why prose cannot be **run** by someone who disagrees.

CHAPTER 30

30

Uselessness restated

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- The thesis, not an apology. Part V's qualification restated as a limit rather than a rescue.

CHAPTER 31

31

What is not claimed

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- The negative inventory, assembled from Appendix C.

CHAPTER 32

32

Kant's moon

NOT YET DRAFTED

This chapter is specified in RFC UCAL-A1 §16.2 and has not been written. Its outline is below so that the book's shape is visible and its length is not quietly understated.

Specified content

- Transcendental illusion is not error but a natural appearance that persists after diagnosis — the astronomer knows the moon is not larger at the horizon and still sees it that way.
- Every reader who sees a 61-digit integer will read it as a fact about being, and no rule prevents that.
- UCAL-E0025 does not cure the illusion; it refuses to compute with it, which is the only thing a specification can do.

AFTERWORD

About the author

Vladimir Ulogov has spent decades building infrastructure for distributed systems — the kind of software that watches other software. Early in his career he worked on monitoring and telemetry platforms; later years took him into federated observability, telemetry buses, and the architecture of systems that have to make sense of millions of data points without losing the thread.

Observability, in the end, is a discipline of **coherence** — of never reporting a state the system cannot account for, of insisting that every signal follow from something real. It is not an accident that a calendar built to point at an origin it cannot measure carries the same instinct: it declares what it knows, declares what it has merely stipulated, and refuses to let the second quietly become the first.

What makes him slightly unusual in his corner of the industry is a tendency to write his own tools — not small utilities, but programming languages. The Bund language (its compiler, its VM, its document store, its parser) lives in a long series of Rust crates on crates.io. `rust_dynamic`, `rust_multistackvm`, `bundcore` — each is a building block that exists because the off-the-shelf options didn't fit the shape of the work. `ucal` grew the same way, from an irritation about units that no existing library was going to fix, because no existing library thought it was a problem.

A work of love

`ucal` is open source, under a permissive licence — you can read it, fork it, study it, modify it, and pass it on. It carries no analytics and no telemetry. It was not built to be sold and it was not built to be adopted; it was built because the question would not go away, and because the only way to find out whether a distinction can be enforced by a compiler is to try to enforce one.

A note on cooperation

Vladimir believes firmly in the human capacity for mutual help — that we make better work, and live better lives, when we share what we know and what we build. Open source is one of the most concrete expressions of cooperation our era has produced: code read, improved, and passed forward without payment, without permission, by people who will never meet.

This book is offered in the same spirit. If it helps you see one distinction more clearly — between what an instrument measures and what it merely points at — that is enough.

Where to find more

GitHub `@vulogov` — the source for `ucal`, `Bund`, and the dozen-plus Rust crates that carry the infrastructure. Issues and pull requests welcome.

LinkedIn `/in/vladimirulogov` — posts on observability, the occasional long-form essay.

YouTube `@vulogov` — talks and walkthroughs from the conference trail.

If you find a claim in this book that the source tree does not support, open an issue. That is the only kind of correction the book is built to receive, and it is the kind it most wants.

AN INSTRUMENT FOR THE IMMEASURABLE

END OF THE BOOK